

Module-6-ML

Module 6

Machine Learning Giralyn R Ongco PhD TM ■ Machine Learning (ML) is a field of computer science that enables systems to learn from data and improve performance without being explicitly programmed Introduction to Machine Machine Learning is a subset of artificial intelligence that focuses on building systems that can: Learning □ Learn from data □ Identify patterns □ Make predictions or decisions Giralyn R Ongco PhD TM Machine learning usually follows these steps:

2. Training

How Machine Learning Works

1. Data Collection: 2. Training 3. Prediction

The system gathers The computer analyzes After learning patterns, data such as: the data to find patterns. Giralyn R Ongco PhD TM Data The foundation of ML.

- images predictions on new data.
- text Example:
- numbers The system may discover: Example:
- audio • Students with high The system can predict:
- sensor readings attendance often get •Which students may
- student attendance •Which email is spam
- quiz scores
- assignment grades

1.Provide the model with input data (features)

Input (X): Study Hours

2.Provide the corresponding correct output (labels)

Output (Y): GA

4.The model can then predict outputs for new data

The model learns how study hours affect GA and can predict GA for a new student Study GA Hours 5 2.8 8 3.2 Giralyn R Ongco PhD TM Types of Supervised Learning 1■■■Regression •Predicts continuous values •Example: Predict GA, salary, temperature 2■■■ Classification •Predicts

categories •Example: Giralyn R Ongco PhD TM Supervised vs Unsupervised Learning Unsupervised Learning Unsupervised learning is a type of machine learning where the model is trained using unlabeled data, meaning there are no predefined answers or There are no labels like outputs. How It Works The model groups them into 1.Provide the model with input data only (no labels) clusters: 2.The model analyzes the data

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- Pass / Fail
- Spam / Not Spam

4.Outputs are clusters or structures, not exact answers

Equation of Linear Regression Study GA ■ = ■■ + ■ Hours Where: •y = predicted value (e.g., GPA) 5 2.8 •x = input variable (e.g., study hours) •m = slope (rate of change) 8 3.2 •b = intercept (starting point Interpretation •If m is positive → as x increases, y increases •If m is negative → as x increases, y decreases Example: •More study hours → higher GPA Giralyn R Ongco PhD TM Classification (Logistic Regression) Classification is used to predict categories or classes. Logistic Regression is a classification algorithm that predicts outcomes like: Study Hours Result •Yes / No •Pass / Fail 3 Fail •Fraud / Not Fraud 6 Pass Output Instead of predicting numbers, it predicts probability: •0 → Fail •1 → Pass Giralyn R Ongco PhD TM

- 3.7
- 3.8
- 3.6

YOU

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